Tutorial 8

The core exercises are considered part of the course material, and you are advised to at least finish these. For additional practice and interesting applications, advanced exercises are available as well – feel free to pick and choose from these.

Unless otherwise specified, we expect you to create your own algorithms for the exercises in each tutorial. This means that you should not use libraries (e.g. SciPy) for the methods that are part of the course material.

Core exercises

1. **General least-squares estimates**
   In this exercise we will use the Levenberg-Marquardt method in order to fit several different models.
   (a) Let's consider that we have a model that follows the following curve:
   $$f(x) = \frac{a}{x} + bx + c \tag{1}$$
   (b) Generate 1,000 realizations of this curve with $(a = 2, b = 1, c = 2)$, for 20 points evenly spaced between 0.5 and 4. On top of the curve add white Gaussian noise with $\sigma = 2$ and use the Levenberg-Marquardt method to find the a, b, c parameters.
   (c) Plot the realizations, the model and the best fit you obtained.
   (d) Repeat the exercise with a Gaussian noise of $\sigma = 1$, but instead of only using 20 evenly spaced points use 10, 20 and 30 evenly spaced points and compare how well we still can retrieve the model.

2. **Galaxy stellar mass function**
   The galaxy stellar mass function describes how many galaxies there are with a certain stellar mass, per unit volume. In this exercise we will use a data set with the observed stellar mass values of galaxies for 5 galaxy clusters at redshift 0.101. The dataset can be downloaded from `https://home.strw.leidenuniv.nl/~daalen/NURfiles/smfdata.txt`, and it contains the stellar mass (m_*) in $\log_{10}$ units.
   (a) Construct the stellar mass function (SMF) for this data, which is the number of galaxies per mass bin per cluster. Use a convenient number of bins for $\log_{10} m_*$ range [8.5, 12].
   (b) The stellar mass function can be represented by the Schechter function as:
   $$\Phi\left(\text{cluster}^{-1}\text{dex}^{-1}\right) = \ln(10) \cdot \Phi^* \cdot \left(10^{(M-M^*)(1+\alpha)}\right) \exp\left(-10^{(M-M^*)}\right) \tag{2}$$
   where $M \equiv \log_{10}(m_*)$. The parameters $M^* \equiv \log_{10}(m_*^*)$ is the characteristic mass, α is the low-mass slope, and Φ^* is the overall normalisation. Fit the data to obtain these parameters. Think about how to do the fitting! Plot your fitted line along with the SMF and your fitted parameters.

Advanced exercises

3. **Non-linear fitting**
   In astrophysics we often have equations that we want to fit that are non-linear in nature. In this exercise we will look at different equations that are important in the case of emission from galaxies, stars or the ISM. Using fitting methods discussed in the lecture try to fit the different equations to your own created data.

(a) Active galactic nuclei often have radio emission that follows a power-law profile with an exponential cut-off given by:

$$f(x) = A\left(\frac{x}{x_0}\right)^{-\alpha} \exp\left(-\frac{x}{x_{\text{cut-off}}}\right). \tag{3}$$

Consider values of $A = 10$, $x_0 = 10$ and $x_{\text{cut-off}} = 100$ and $\alpha = 1.5$. Generate 10 points between 1 and 200. Add to the points white Gaussian noise with $\sigma = 0.5$ dex. Calculate the best estimate for 1000 realizations. Repeat your procedure for white Gaussian noise with $\sigma = 1.0$ dex and 0.1 dex. Plot the inferred distribution of parameters for the power law with exponential cut-off.

(b) Many objects in astrophysics have broken power laws, an example of this are gamma-ray bursts (GRBs) that have several power law breaks in their spectrum. Often the power law will go smoothly from one index to the other. In general a smoothly broken power law is given by:

$$f(x) = A\left(\frac{x}{x_b}\right)^{-\alpha_1}\left(\frac{1}{2}\left[1+\left(\frac{x}{x_b}\right)^{1/\Delta}\right]\right)^{(\alpha_1-\alpha_2)\Delta}. \tag{4}$$

Consider the case that $A = 10$, $\Delta = \frac{1}{10}$, $\alpha_1 = 1$, $\alpha_2 = 2$ and $x_b = 50$, generate 10 points between 1 and 200. Add to the points white Gaussian noise with $\sigma = 0.5$ dex. Calculate the best estimate for 1000 realizations. Repeat your procedure for white Gaussian noise with $\sigma = 1.0$ dex and 0.1 dex. Plot the inferred distribution of parameters for the broken power law.

(c) One of the most known emission mechanisms is black body radiation, black body radiation produces a Planck curve, which has the functional form of:

$$f(x) = A\frac{x^3}{e^{\alpha\cdot x}-1}. \tag{5}$$

Consider this curve with $A = 10$ and $\alpha = 1$, generate 10000 realizations with 10 random points between 0 and 12 (so the points vary too between the different realizations), use a white Gaussian noise of 4 and plot the distribution of A and α. Repeat this analysis for noise with $\sigma = 2$ and 0.5.